Kaan Aytekin

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EDUCATION

University of Rochester **Rochester, NY**

Bachelor of Science and Master of Science, Mechanical Engineering *May 2027*

- GPA 3.92 out of 4.00, Dean's List all applicable semesters
- Davis UWC Scholar, Tau Beta Pi Scholar, Helmut Weymann Prize for Engineering Experimentation

ENGINEERING EXPERIENCE (See Portfolio for more detail - www.kaanaytekin.com)

ASML – Air Bearing Metrology and Testing Rig (Capstone Project) **Rochester, NY**

Team Member *January 2026- May 2026*

- Engineered a precision metrology rig from feasibility, concept to product to characterize load capacity and micron fly-height of commercial air bearings, enabling testing of cost-effective alternatives to custom bearings.
- Conducted Finite Element Analysis (Static, Modal Response) and first principle engineering equations on loading structure to ensure safety during peak loading. Validated results with the requirements and specifications.
- Sourced commercially available air bearings based on business and engineering requirements, tested the most competitive ones with the metrology rig to recommend cost-effective alternatives to the current air bearings.

UR Baja SAE (Offroad Car Collegiate Design Series) **Rochester, NY**

Chief Engineer and Drivetrain Project Team Co-Leader *May 2023 – May 2025*

- Led a 30-member multi-disciplinary team through complete vehicle mechanical design and manufacture, achieving a 40% increase in national competition rankings while reducing the project spending by 5%.
- Designed the team's first-ever custom differential along with a custom transfer case and driveshaft using NX.
- Validated gear-train integrity by applying AGMA standards and MATLAB to create solvers for bevel gear forces.
- Optimized axle and universal joint geometry using engineering analysis methods such as FEA (NASTRAN), dynamics simulations (Siemens Simcenter 3D), solid mechanics principles and Design for Manufacturing (DFM), troubleshooting critical interference and loosening issues while achieving a 64% reduction in material costs.
- Manufactured 50+ custom designed axle yokes in a 4-axis CNC mill, used a manual mill to manufacture axle flanges, inserts and more. Designed custom fixture and tooling for sheet metal bending.

Newtonian Telescope Optomechanical Design and Manufacture **Rochester, NY**

Optomechanical Advisor to Optics Capstone Project *January 2026- May 2026*

- Owned the mechanical architecture, precision design, and CNC manufacturing of a telescope, engineering mirror mounts and optical alignment mechanisms capable of 0.015-degree tilt increments. Identified and documented design requirements and specifications through transdisciplinary communication with optical engineers.
- Performed FEA on primary mirror to model the deflection due to gravity at various telescope angles, fitting surface deformation data with Zernike polynomials to predict and mitigate line-of-sight and alignment errors.

Laboratory for Laser Energetics (DoE funding $100M+, 400+ employees) **Rochester, NY**

Research Assistant *March 2023 - Present*

- Conducted in-depth numerical calculations of free-free laser absorption rates in plasmas, incorporating quantum scattering theory and screening effects. Hands on experience in optical experiment setup.
- Performed computations of 100 absorption correction tables (Gaunt Factor) using a Linux-based HPC cluster. Fitted a parametric analytical expression with a three-variable equation to extract data for integration into opacity and hydro-codes along with fusion experiments. Leveraging Python skills (NumPy, SciPy and Matplotlib).

Torsion Testing Rig for 4130 Steel Axles **Rochester, NY**

Author *January 2024 - May 2024*

- Developed a custom torque testing rig and testing methodology to measure the torsional yield characteristics of heat-treated 4130 steel (no-treatment, annealed and quenched samples) using strain gauges, NI data acquisition devices, analyzing data using MATLAB and first principles engineering equations.
- Conclusively validated that targeted heat treatment statistically and significantly improves the torsional yield point of steel half-shaft connections, supporting component weight reduction in vehicle design.

AWARDS

APS DPP '24 Outstanding Poster Award **Atlanta, GA**

American Physical Society Division of Plasma Physics Annual Meeting *October 2024*

- Awarded to top 5 student presenters out of approx. 150 presenters based on technical writing and presentation.

SKILLS SUMMARY & INTERESTS

- **CAD:** Siemens NX, Solidworks. **Simulation/CAE:** Simcenter 3D, NASTRAN (linear, nonlinear, real eigenvalues), Multibody Dynamics, ANSYS Fluent (Workbench, Dynamic Meshing, VoF Mixtures) **Data Analysis/Coding:** MATLAB (Real Time Data Acquisition, NI, Simulink, Strain Gages), Python (NumPy, SciPy, Matplotlib, Large dataset analysis) **Other:** Opto-Mechanical Design, Lathe, Mill, CNC machining, Prototyping, 3D printing.
- Turkish (Native), English (Fluent), German (B1)
- **Music,** played and recorded piano and guitar, led band as the club leader and the lead guitarist.